Algebraic solutions

1

a $x = \frac{\pi}{2}, -\pi$

b $x = \frac{\pi}{2}, \pi$

c $x = \frac{\pi}{30}, \frac{11\pi}{30}, \frac{61\pi}{30}, \frac{71\pi}{30}$

d $x = \frac{\pi}{12}, \frac{7\pi}{12}, -\frac{23\pi}{12}, -\frac{17\pi}{12}$

e $x = \frac{2\pi}{15}, \frac{17\pi}{15}$

f $x = \frac{11\pi}{18}, -\frac{\pi}{18}, -\frac{13\pi}{18}$

g $x = -\frac{\pi}{15}, \frac{19\pi}{15}, \frac{59\pi}{15}, -\frac{41\pi}{15}$

h $x = \frac{\pi}{2}, \pi$

i $x = \frac{5}{12}, \frac{17}{12}, -\frac{19}{12}, -\frac{7}{12}$

j $x = -\frac{\pi}{120}, \frac{29\pi}{120}, \frac{59\pi}{120}, -\frac{31\pi}{120}$

k $x = -\frac{\pi}{9}, \frac{\pi}{9}$

l $x = -\frac{5\pi}{18}, -\frac{\pi}{18}, \frac{7\pi}{18}$

m $x = \frac{13\pi}{36}, \frac{31\pi}{36}$

n $x = 0, \frac{4\pi}{9}, \frac{2\pi}{3}$

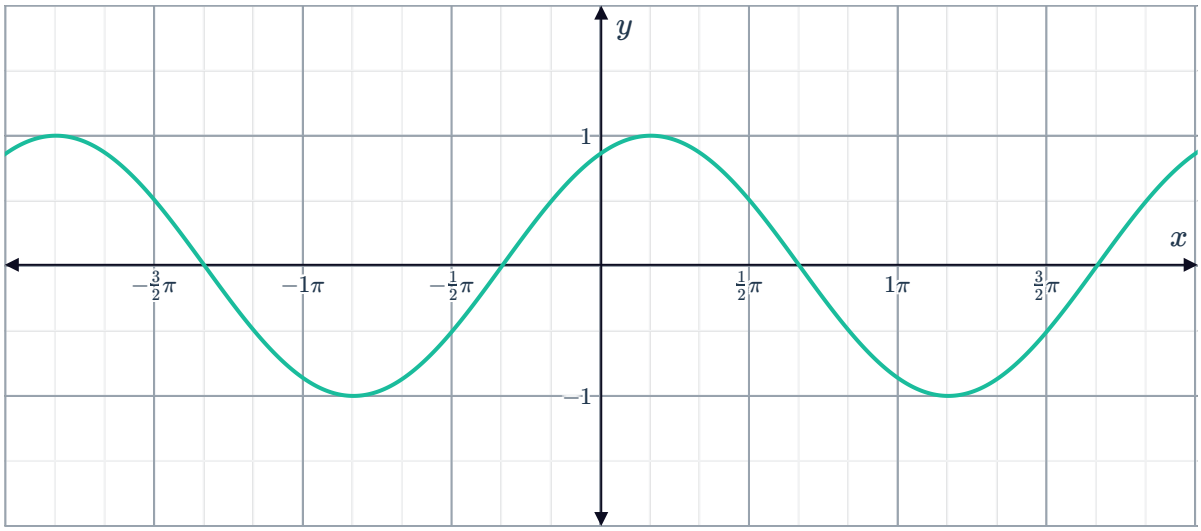
o $x = \frac{3\pi}{4}, \frac{7\pi}{4}$

p $x = \frac{\pi}{4}, \frac{5\pi}{12}, \frac{3\pi}{4}, \frac{11\pi}{12}$

Graphical solutions

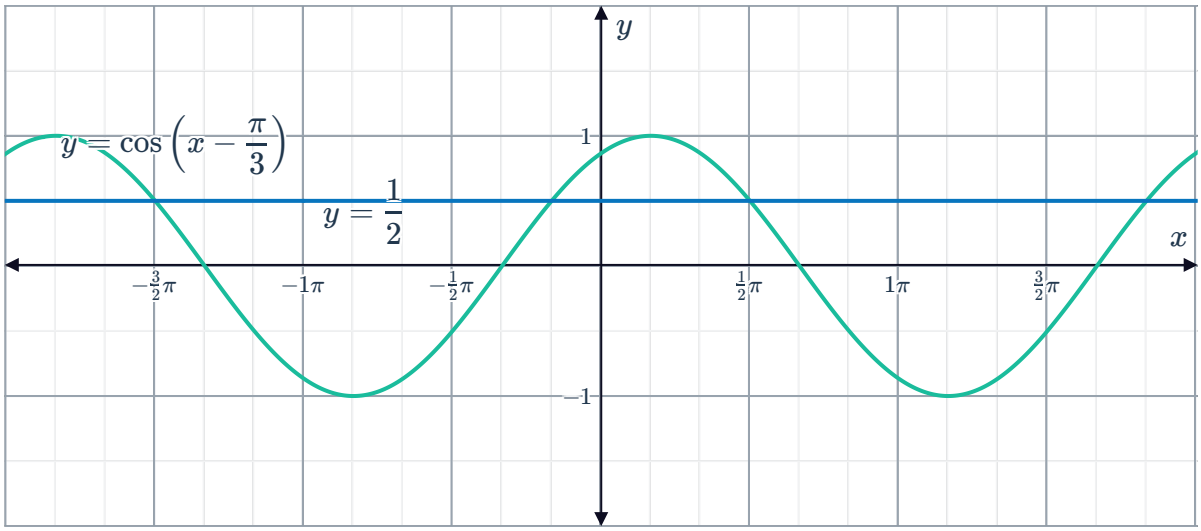
2

a



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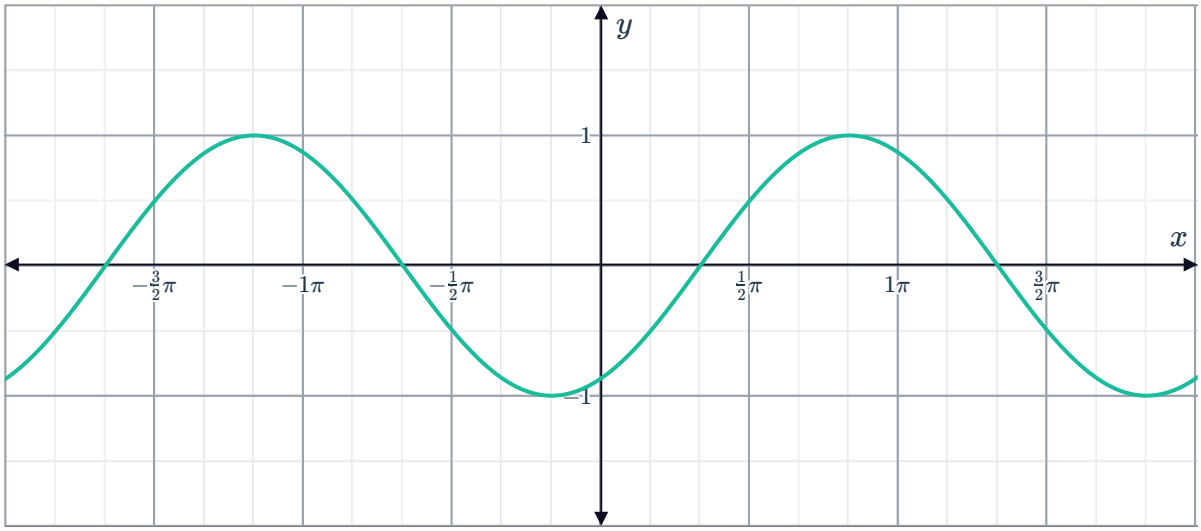
b



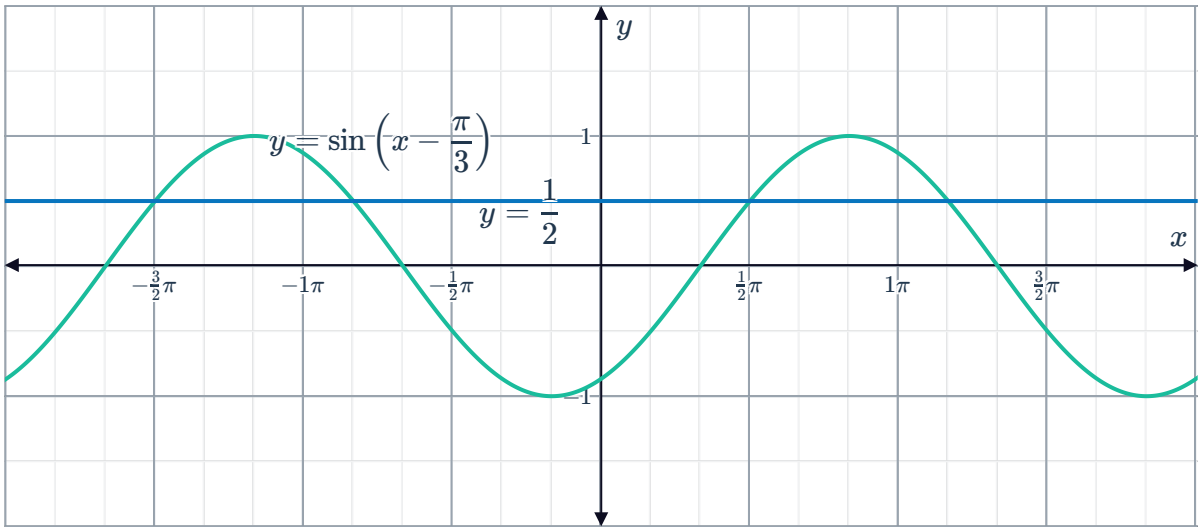
c $x = -\frac{3\pi}{2}, -\frac{\pi}{6}, \frac{\pi}{2}, \frac{11\pi}{6}$

3

a



b

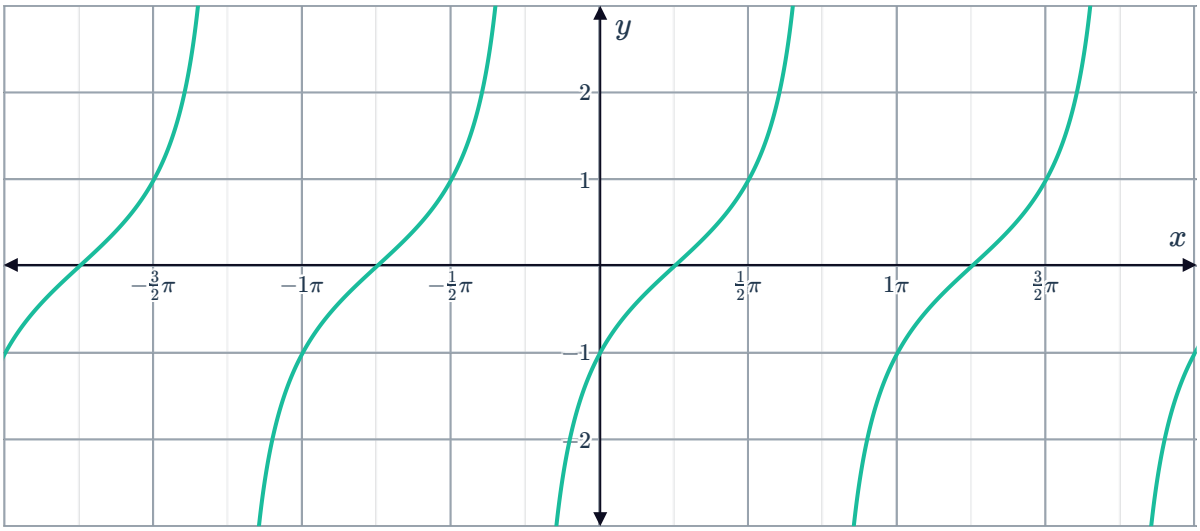


c $x = -\frac{3\pi}{2}, -\frac{5\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}$

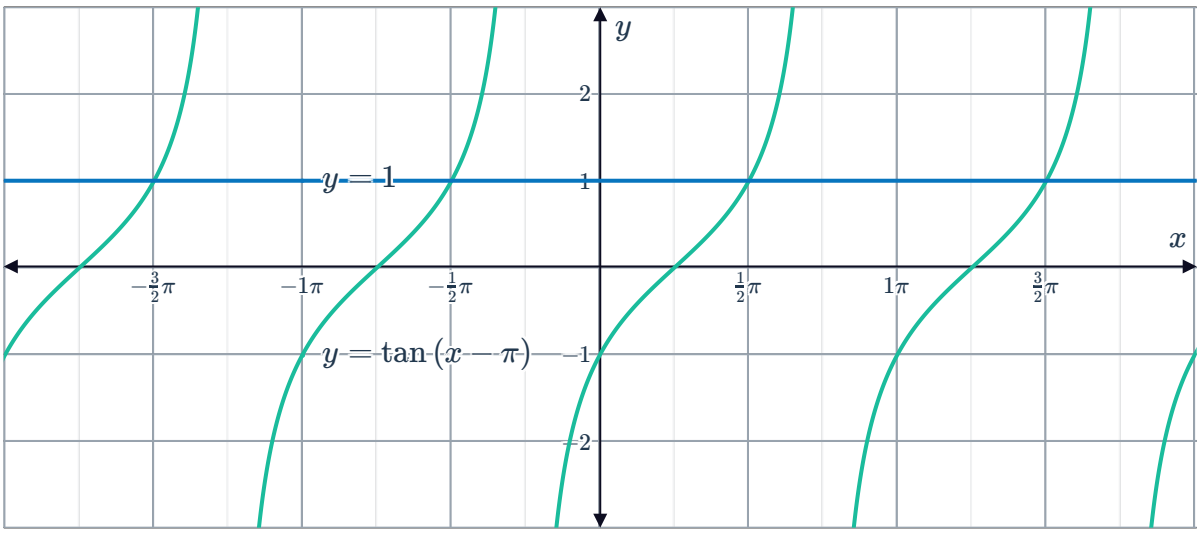
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4

a



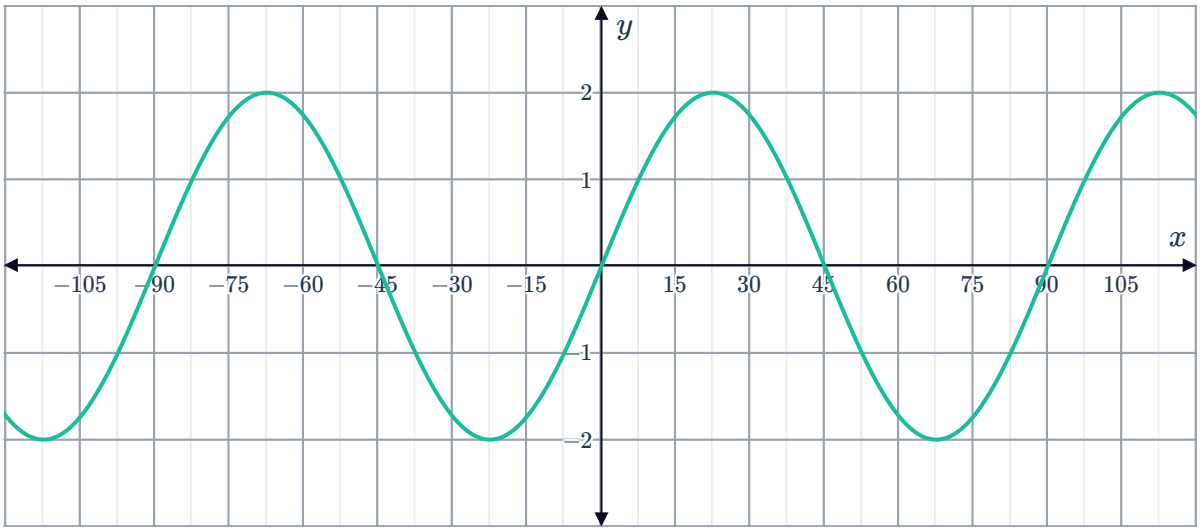
b



c $x = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

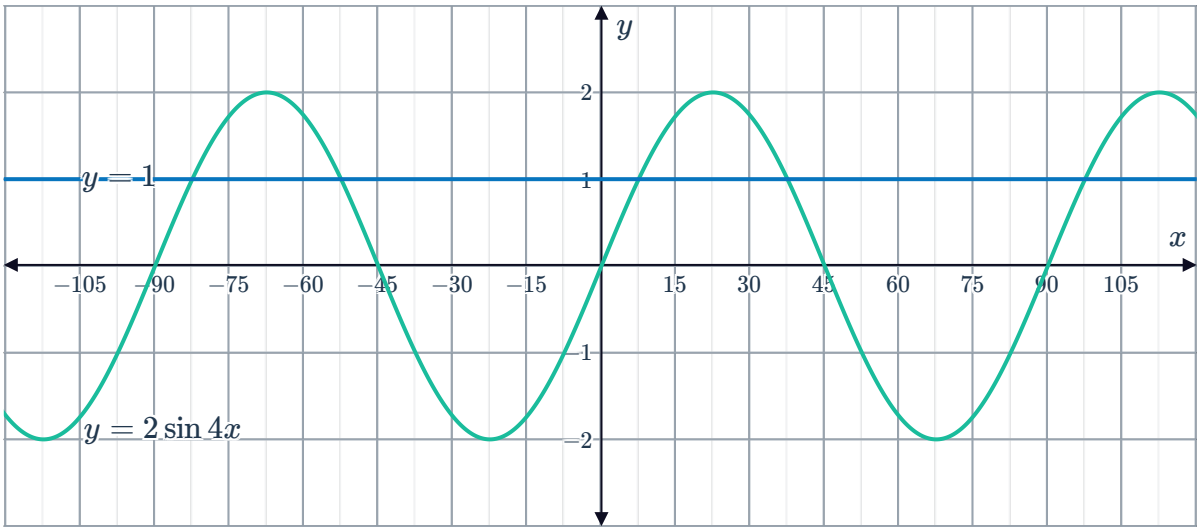
5

a



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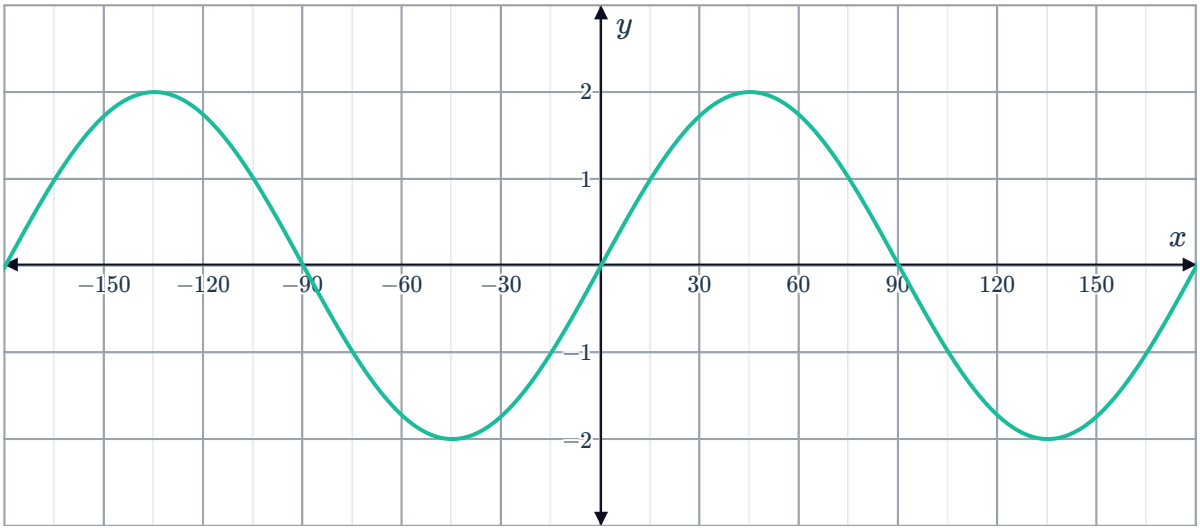
b



c $x = -82.5^\circ, -52.5^\circ, 7.5^\circ, 37.5^\circ$

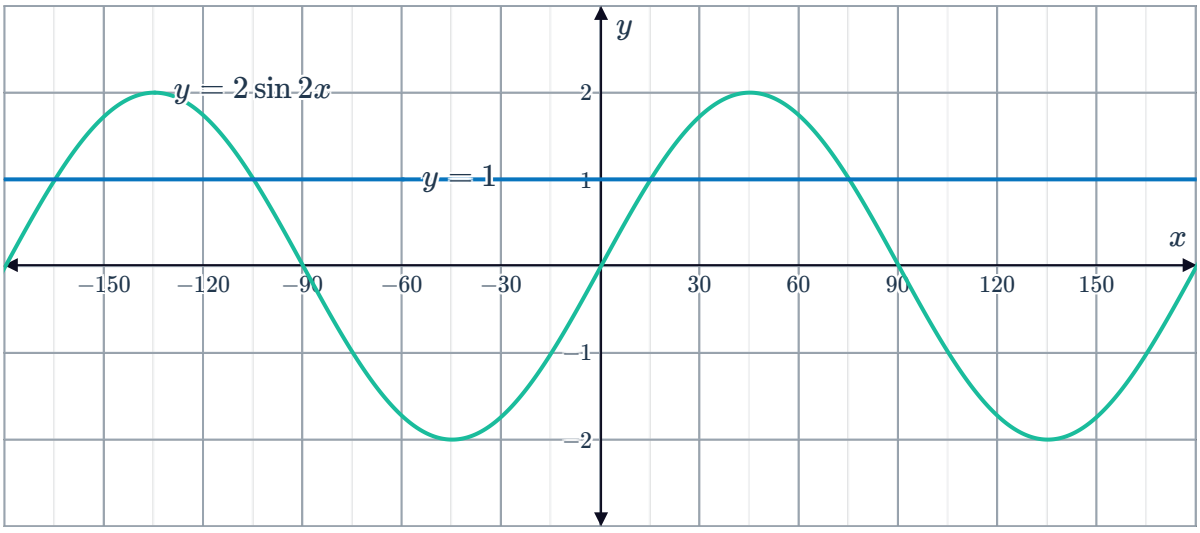
6

a



b $y = 1$

c

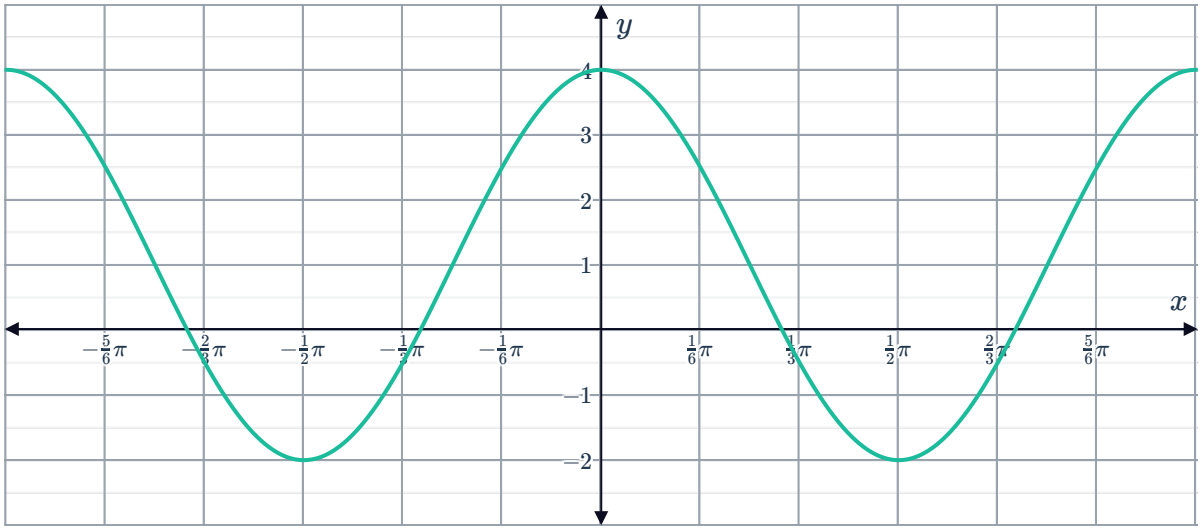


d $x = -165^\circ, -105^\circ, 15^\circ, 75^\circ$

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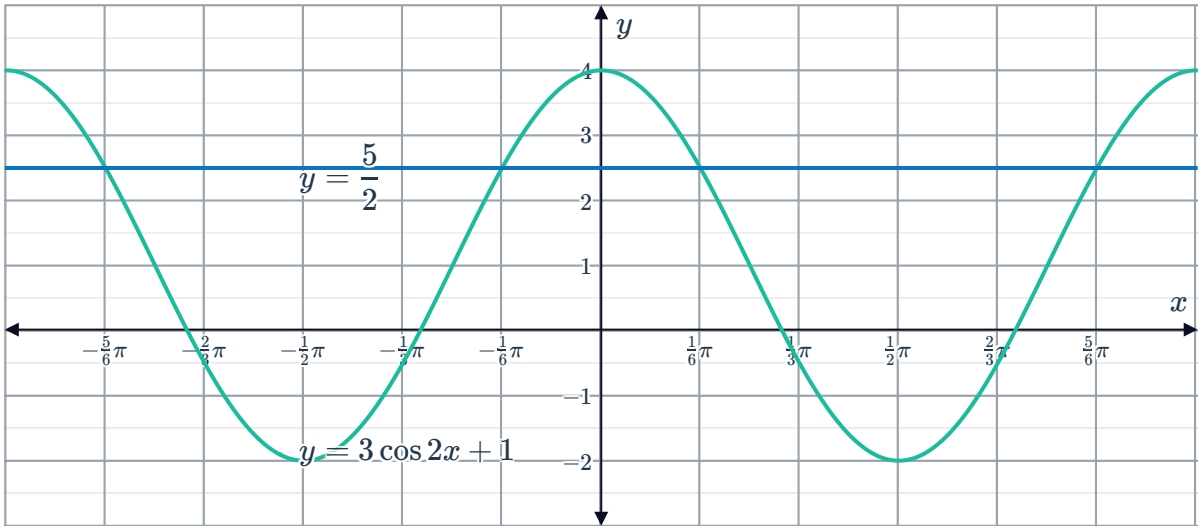
7

a



b $y = \frac{5}{2}$

c

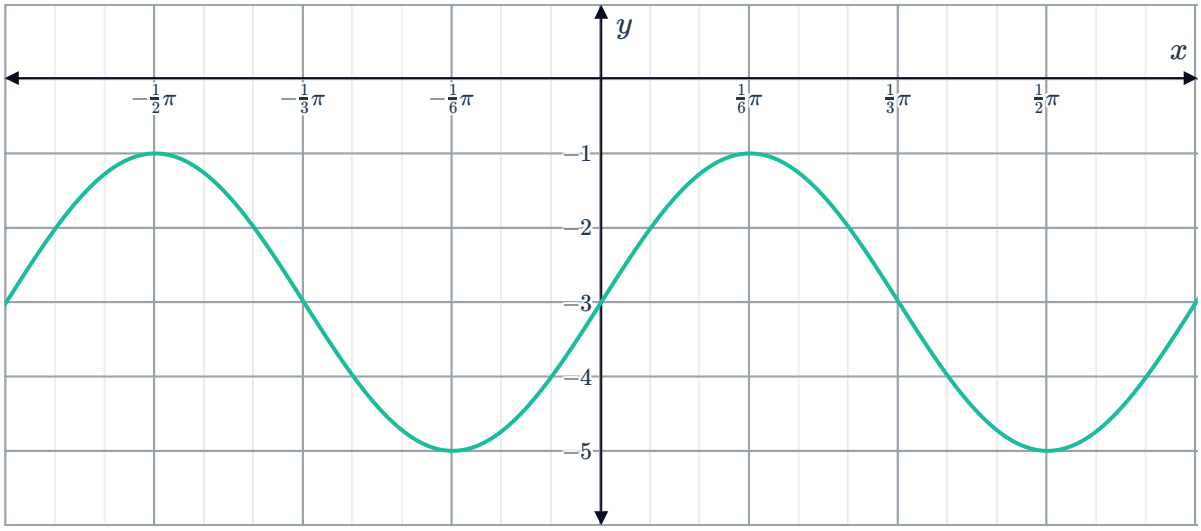


d $x = -\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$

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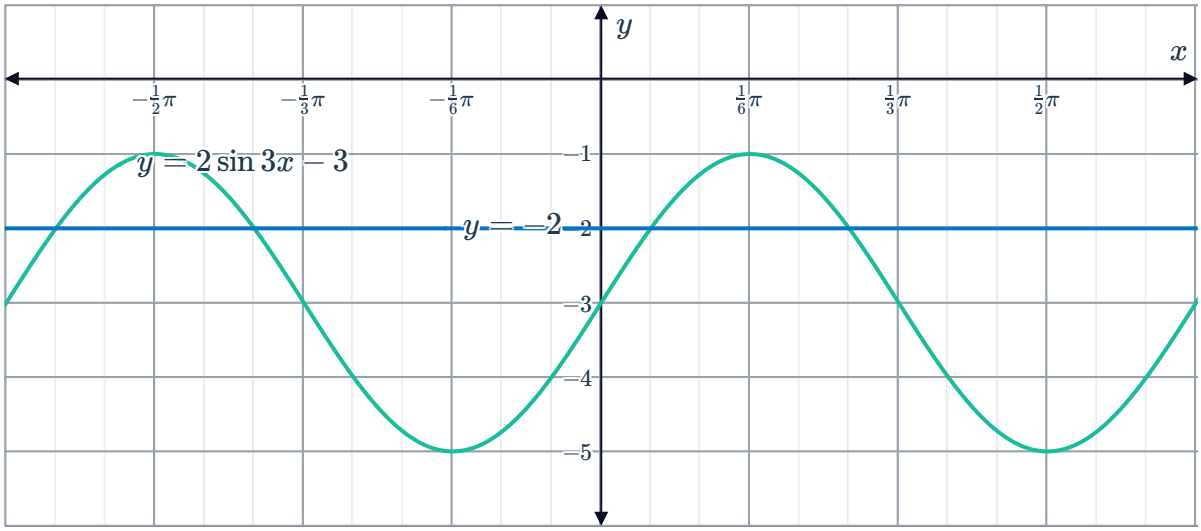
8

a



b $y = -2$

c

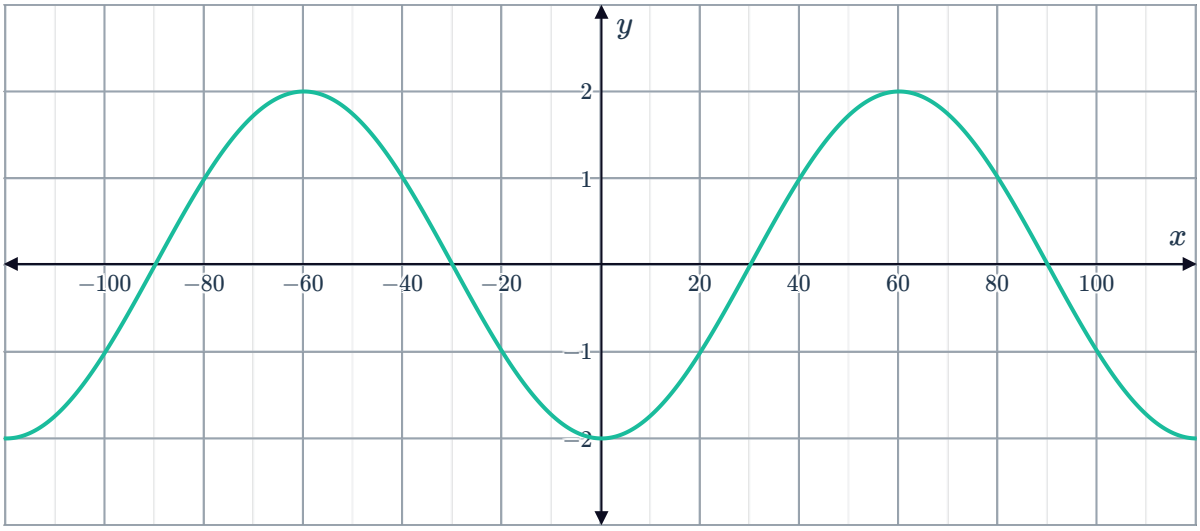


d $x = -\frac{11\pi}{18}, -\frac{7\pi}{18}, \frac{\pi}{18}, \frac{5\pi}{18}$

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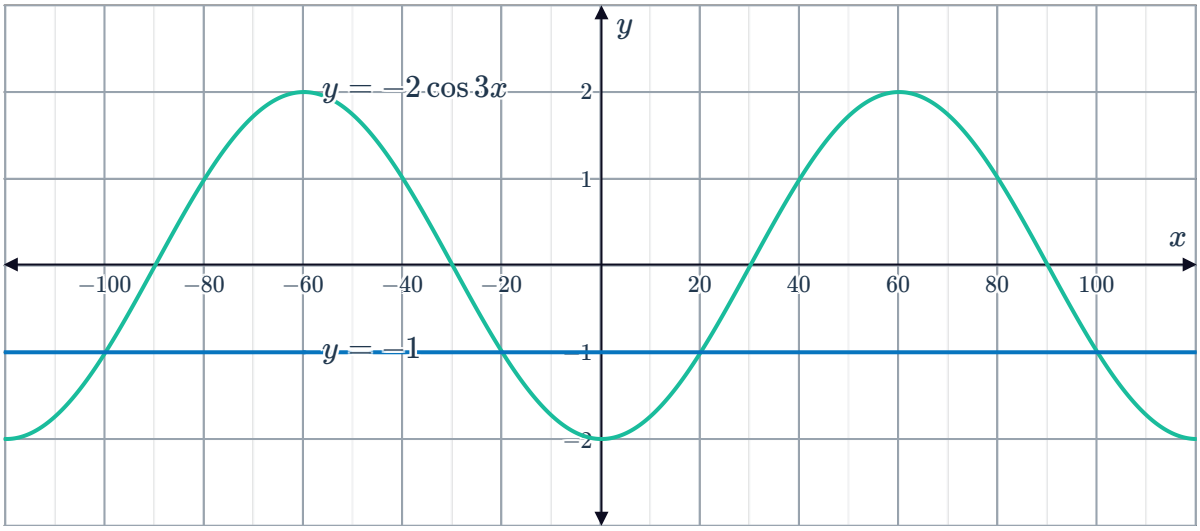
9

a



b $y = -1$

c



d $x = -100^\circ, -20^\circ, 20^\circ, 100^\circ$

Technology

10

a $x = 0.23, 2.92, 3.37, 6.06$

b $x = 0.21, 1.37$

c $x = -3.54, 2.75, 9.03$

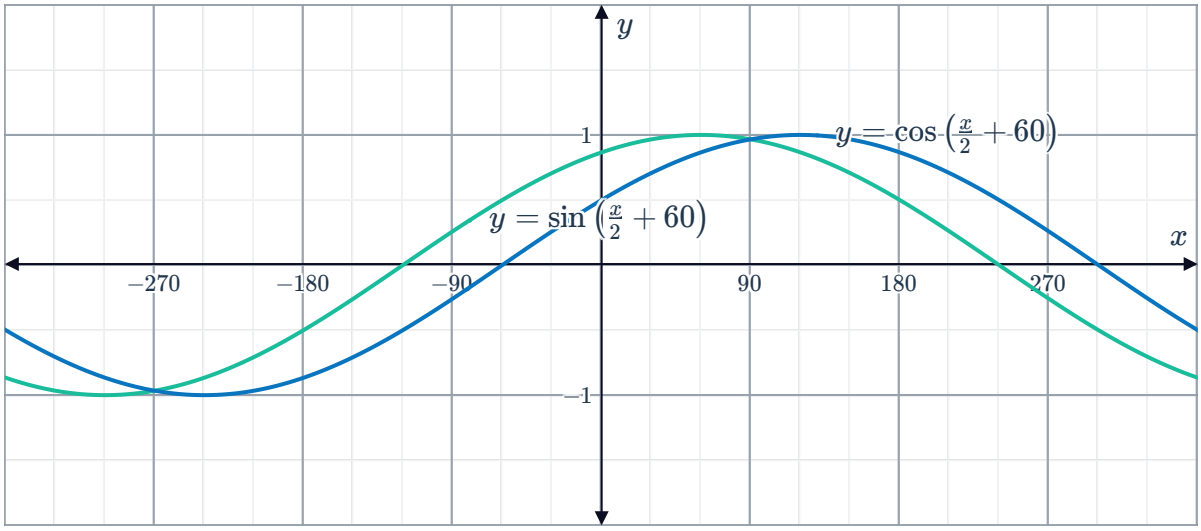
d $x = 8.90$

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11

a $y = \sin\left(\frac{x}{2} + 60^\circ\right)$, $y = \cos\left(\frac{x}{2} - 60^\circ\right)$

b



c $x = -270^\circ, 90^\circ$

12

a $x = 0^\circ, 102^\circ, 180^\circ, 258^\circ$

b $x = 101^\circ, 303^\circ$

c $x = 35^\circ, 215^\circ$

d $x = 65^\circ, 142^\circ, 218^\circ, 295^\circ$

e $x = 7^\circ, 85^\circ, 180^\circ, 275^\circ, 353^\circ$

f $x = 29^\circ, 92^\circ, 147^\circ, 213^\circ, 268^\circ, 331^\circ$

g $x = 180^\circ, 211^\circ, 304^\circ$

h $x = 133^\circ, 227^\circ$

i $x = 42^\circ$

j $x = 54^\circ$

13

a $x = 1.765, 5.280$

b $x = 0, 0.318, 3.142, 4.965, 6.283$

c $x = -1.377$

d $x = 0.655, 2.007, 4.277, 5.628$

e $x = 0.608, 3.749$

f $x = 0.126, 1.478, 3.142, 4.805, 6.157$

g $x = 0.507, 1.619, 2.554, 3.729, 4.664, 5.776$

h $x = 0.840, 3.474$

i $x = 3.142, 3.680, 5.307$

j $x = 2.426, 3.857$

k $x = 1.760$

l $x = 1.110$

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